Exact Observation-Time Stabilization Laws for the Brownian Factor-of-IID Sampler

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Abstract

OpenAI's Brownian factor-of-IID sampler transforms independent Brownian innovations into a path field whose joint law with its binary output X is that of

$$Y_i(t) = tX_i + B_i(t),$$

where the B_i are independent standard Brownian motions independent of X. We analyze the continuous-time midpoint readout $\widehat{X}_i(t) = \mathbf{1}\{Y_i(t) > t/2\}$. If T_i is the supremum of the times at which coordinate i is decoded incorrectly, then, conditionally on the entire sampled configuration, the variables T_i are independent with common law $4\chi_1^2 = \mathrm{Gamma}(1/2, 8)$. In particular, the stabilization-time field is independent of the sampled configuration. We obtain the exact finite-set law

$$\mathbb{P}\Big(\max_{i\in A} T_i \le t\Big) = \left[1 - 2\Phi\left(-\frac{\sqrt{t}}{2}\right)\right]^{|A|},$$

an exact binomial law for the remaining terminal defects, explicit confidence horizons, and a Gumbel limit with centering $8\log M - 4\log\log M - 4\log\pi$ and scale 8. We also prove moment asymptotics, unequal-rate and deterministic-clock versions, and a quantitative Poisson transfer principle for heterogeneous coordinates. For finite target laws we also identify the sampler drift through the vector I–MMSE identity, obtaining an exact entropy-dissipation formula. The results apply, in particular, to the free uniform spanning forest sampler constructed in the OpenAI preprint.

1 Introduction

The free uniform spanning forest is a central probability measure on infinite graphs; see, for example, [BLPS01, LP16]. A recent OpenAI preprint proves that the free uniform spanning forest on every infinite, connected, locally finite, simple, unweighted graph is a factor of independent vertex labels [Ope26]. A principal ingredient is a Brownian sampler for binary processes satisfying a uniform response bound. Its output is recovered from the asymptotic slopes of a deterministically reconstructed observation field.

The sampler proof contains a stronger path-law statement than is needed merely to identify the final output. If W denotes the independent Brownian input, $Z(W)$ the reconstructed path field, and $\Psi(W)$ the binary output, then

$$\big(Z(W), \Psi(W)\big) \overset{\mathrm{d}}{=} \big((tX_i + B_i(t))_{i,t}, X\big), \tag{1.1}$$

where X has the target law and the Brownian motions B_i are independent of one another and of X. We take (1.1) as the sole input from the sampler construction and determine exactly how quickly the natural midpoint readout stabilizes.

The distinction between a snapshot error and a future error is important. At a fixed time t, one coordinate is decoded incorrectly with probability $\Phi(-\sqrt{t}/2)$. The probability that it makes an error at some time after t is twice as large:

$$\mathbb{P}(T_i > t) = 2\Phi\left(-\frac{\sqrt{t}}{2}\right). \tag{1.2}$$

The second half of this probability comes from paths that are currently on the correct side of the threshold but later return to it. Thus T_i is a last-passage variable rather than a stopping time.

Our main theorem identifies the entire marked law:

$$\mathcal{L}(X, (T_i)_{i\in I}) = \mathcal{L}(X) \otimes \bigotimes_{i\in I} \mathrm{Gamma}\left(\frac{1}{2}, 8\right). \tag{1.3}$$

Arbitrary dependence in the target process disappears from the stabilization marks. The reason is the midpoint threshold: after a sign change determined by X_i, either target bit produces the same Brownian motion with drift $-1/2$.

The proof of the stabilization laws is self-contained after (1.1). We record a Brownian last-passage lemma, derive the product marking, and then develop exact finite-size, extreme-value, and heterogeneous-clock consequences. For finite target laws we additionally use the posterior-mean interpretation of the sampler drift and the vector I–MMSE identity to record an exact entropy-dissipation formula. Standard background on Brownian hitting and last-passage identities may be found in [KS91, RY99, BS02]; background on extreme-value limits may be found in [Res87].

The stabilization parameter studied here is Brownian observation time after the path field $Z(W)$ has been reconstructed. It is not a coding radius in the underlying IID vertex labels, and none of the results below asserts that the factor map is finitary. The exact law instead quantifies the online readout of the reconstructed observation field.

Provenance

The deterministic inversion, innovation representation, posterior-drift identification, and path-law identity (1.1) are due to the OpenAI preprint [Ope26], cited at the fixed source commit recorded in the bibliography. The stabilization theorems and their consequences below are derived from the path-law identity; the entropy formula also uses the posterior-drift identification.

2 The sampler representation

Let I be a finite or countable index set. We isolate the consequence of the OpenAI sampler that is used throughout the paper.

Proposition 2.1 (OpenAI path representation)**.** *Suppose that a probability law ν on $\{0,1\}^I$ satisfies the hypotheses of the Brownian sampling proposition in [Ope26, Section 4]. Let $W = (W_i)_{i\in I}$ be independent standard Brownian paths, let $Z(W)$ be the Borel solution of the sampler's deterministic inverse equation*

$$Z_i(W)(t) = W_i(t) + \int_0^t b_i(s, Z(W)(s))\, \mathrm{d}s, \tag{2.1}$$

and let $\Psi(W)$ be its binary slope readout. There exist a random configuration $X \sim \nu$ and independent standard Brownian motions $(B_i)_{i\in I}$, independent of X, such that

$$\big(Z(W), \Psi(W)\big) \overset{\mathrm{d}}{=} \big((tX_i + B_i(t))_{i\in I,\, t\geq 0}, X\big). \tag{2.2}$$

Proof. The cited proposition proves that independent Brownian input is transformed by Z into the complete joint path law of the reference observations $Y_i(t) = tX_i + B_i(t)$. Its slope readout is a Borel function of the path field, and, in the reference model, that readout equals X almost surely. Applying the same Borel readout to two path fields with the same law gives (2.2). See the proof of the proposition entitled "Sampling from independent Brownian paths" in [Ope26, Section 4]. □

For the free uniform spanning forest, the OpenAI preprint verifies the required uniform response estimate with Lipschitz constant $1/2$ [Ope26, Section 4]. Nothing in our argument below otherwise uses spanning-forest structure.

In view of Proposition 2.1, it is enough to work in the reference model

$$Y_i(t) = tX_i + B_i(t), \tag{2.3}$$

where X has an arbitrary law on $\{0,1\}^I$ and the B_i are independent of X.

Definition 2.2 (Midpoint readout and last error)**.** For $t \geq 0$, define

$$\widehat{X}_i(t) = \mathbf{1}\{Y_i(t) > t/2\}. \tag{2.4}$$

The last-error time at coordinate i is

$$T_i = \sup\{s \geq 0 : \widehat{X}_i(s) \neq X_i\}. \tag{2.5}$$

For a finite $A \subseteq I$, put

$$T_A = \max_{i\in A} T_i. \tag{2.6}$$

The supremum convention in (2.5) is deliberate. At a final threshold contact, the strict readout in (2.4) can be correct for target bit 0 and incorrect for target bit 1. In either case, all times strictly greater than T_i are decoded correctly.

3 Posterior drift and entropy dissipation

The path representation also gives an exact information identity when the index set is finite. Entropies and mutual informations in this section use natural logarithms.

Theorem 3.1 (Entropy–MMSE identity)**.** *Let I be finite, let X have an arbitrary law on $\{0,1\}^I$, and let $Y_i(t) = tX_i + B_i(t)$ as in (2.3). Put*

$$\mathcal{Y}_t = \sigma\{Y_i(s) : i \in I,\ 0 \leq s \leq t\}, \qquad m_i(t) = \mathbb{E}[X_i \mid \mathcal{Y}_t].$$

Then, for every $T \geq 0$,

$$I(X; \mathcal{Y}_T) = \frac{1}{2}\int_0^T \sum_{i\in I} \mathbb{E}\big[(X_i - m_i(t))^2\big]\ \mathrm{d}t. \tag{3.1}$$

Moreover,

$$H(X) = \frac{1}{2}\int_0^\infty \sum_{i\in I} \mathbb{E}\big[(X_i - m_i(t))^2\big]\ \mathrm{d}t = \frac{1}{2}\int_0^\infty \sum_{i\in I} \mathbb{E}\big[m_i(t)(1 - m_i(t))\big]\ \mathrm{d}t. \tag{3.2}$$

Proof. For fixed t, the likelihood of the path $(Y(s))_{0\le s\le t}$ given $X = x$ is, up to an x-independent factor,

$$\exp\!\left(x \cdot Y(t) - \frac{t}{2}|x|^2\right).$$

Thus $Y(t)$ is sufficient for X relative to $\mathcal{Y}_t$. After the normalization $Y(t)/\sqrt{t} = \sqrt{t}\,X + G$, where G is a standard Gaussian vector, the vector I–MMSE identity gives (3.1); see [GSV05]. Since $Y_i(t)/t \to X_i$ almost surely for every i, the full configuration X is measurable with respect to $\mathcal{Y}_\infty = \sigma(\cup_t \mathcal{Y}_t)$. Hence $I(X;\mathcal{Y}_T) \uparrow H(X)$, and monotone convergence in (3.1) proves the first equality in (3.2). The second follows from

$$\mathbb{E}[(X_i - m_i(t))^2 \mid \mathcal{Y}_t] = m_i(t)(1 - m_i(t)).$$

□

Corollary 3.2 (Entropy of a finite sampler output)**.** *Under the hypotheses of Proposition 2.1, suppose I is finite and set $X = \Psi(W)$. Then the sampler drift in (2.1) satisfies*

$$H(\Psi(W)) = \frac{1}{2}\int_0^\infty \sum_{i\in I} \mathbb{E}\Big[\big(\Psi_i(W) - b_i(t, Z(W)(t))\big)^2\Big]\ \mathrm{d}t. \tag{3.3}$$

Proof. In the reference observation model, the OpenAI posterior lemma identifies $b_i(t, Y(t))$ with $\mathbb{E}[X_i \mid \mathcal{Y}_t]$. Transfer this joint law through Proposition 2.1 and apply Theorem 3.1. □

For example, if X is the edge-indicator vector of a uniform spanning tree of a finite connected graph G, then $H(X) = \log \tau(G)$, where $\tau(G)$ is the number of spanning trees. Equation (3.2) gives an exact integral representation of this spanning-tree entropy by the total posterior variance along the Brownian observation channel.

Proposition 3.3 (The Bayes snapshot benchmark)**.** *Under the assumptions of Theorem 3.1, for a coordinate $i \in I$, define the full-field Bayes decision*

$$\widetilde{X}_i(t) = \mathbf{1}\{m_i(t) > 1/2\}.$$

Then, for every $t > 0$,

$$\mathbb{P}(\widetilde{X}_i(t) \neq X_i) \le \mathbb{P}(\widehat{X}_i(t) \neq X_i) = \Phi(-\sqrt{t}/2). \tag{3.4}$$

More precisely,

$$\mathbb{P}(\widehat{X}_i(t) \neq X_i) - \mathbb{P}(\widetilde{X}_i(t) \neq X_i) = \mathbb{E}\Big[|2m_i(t) - 1|\,\mathbf{1}\{\widehat{X}_i(t) \neq \widetilde{X}_i(t)\}\Big]. \tag{3.5}$$

Proof. Condition on $\mathcal{Y}_t$. For a binary variable, choosing the more likely state minimizes conditional error, and the excess conditional risk of the opposite decision is $|2m_i(t) - 1|$. This proves (3.5). The midpoint error probability is the same for both values of X_i and equals $\Phi(-\sqrt{t}/2)$. □

The proposition concerns snapshot risk only. It does not order the last switching times of the midpoint and Bayes decision processes.

4 A Brownian last-passage identity

Write Φ for the standard normal distribution function and $\overline{\Phi} = 1 - \Phi$. Throughout, Gamma$(\alpha, \theta)$ denotes the gamma distribution with shape α and scale θ.

Lemma 4.1 (Exact last-passage law)**.** *Let B be standard Brownian motion and let $a > 0$. Define*

$$L_a^{>} = \sup\{s \geq 0 : B(s) - as > 0\}, \tag{4.1}$$

$$L_a^{\geq} = \sup\{s \geq 0 : B(s) - as \geq 0\}. \tag{4.2}$$

Both variables are finite almost surely and have the same distribution. For $t \geq 0$,

$$\mathbb{P}(L_a^{>} \leq t) = \mathbb{P}(L_a^{\geq} \leq t) = 2\Phi(a\sqrt{t}) - 1. \tag{4.3}$$

Equivalently, if $G \sim N(0,1)$, then

$$L_a^{>} \overset{\mathrm{d}}{=} L_a^{\geq} \overset{\mathrm{d}}{=} \frac{G^2}{a^2}. \tag{4.4}$$

Proof. Set $R(s) = B(s) - as$. Since $B(s)/s \to 0$ almost surely, $R(s) < 0$ for all sufficiently large s, so both last-passage times are finite.

We first recall the hitting probability from a negative starting point. If $R(0) = y < 0$ and $\tau_0 = \inf\{s \geq 0 : R(s) = 0\}$, then

$$\mathbb{P}_y(\tau_0 < \infty) = e^{2ay}. \tag{4.5}$$

Indeed, $e^{2aR(s)}$ is a local martingale. Stopping it on first hitting 0 or $-m$, and then letting $m \to \infty$, gives (4.5). This is the standard one-sided hitting formula for Brownian motion with drift; see, for example, [KS91, Chapter 3].

Fix $t > 0$. Conditional on $R(t) = y < 0$, a future visit to the nonnegative half-line has probability e^{2ay}. If $y > 0$, a future positive value occurs with probability one by continuity. Starting from $y = 0$, the positive half-line is regular for Brownian motion with drift, so a future positive value again occurs with probability one. The distinction at $y = 0$ is immaterial in any case because $R(t)$ has a continuous distribution. The same calculation therefore applies to the strict and non-strict last-passage times:

$$\mathbb{P}(L_a^{\diamond} > t) = \mathbb{P}(R(t) \geq 0) + \mathbb{E}\left[e^{2aR(t)}; R(t) < 0\right], \qquad \diamond \in \{>, \geq\}. \tag{4.6}$$

Now $R(t) \sim N(-at, t)$, so

$$\mathbb{P}(R(t) \geq 0) = \Phi(-a\sqrt{t}). \tag{4.7}$$

Completing the square in the second term gives

$$\begin{aligned}
\mathbb{E}\left[e^{2aR(t)}; R(t) < 0\right] &= \int_{-\infty}^{0} \frac{1}{\sqrt{2\pi t}} \exp\left(2ay - \frac{(y + at)^2}{2t}\right) \mathrm{d}y \\
&= \int_{-\infty}^{0} \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y - at)^2}{2t}\right) \mathrm{d}y \\
&= \Phi(-a\sqrt{t}).
\end{aligned} \tag{4.8}$$

Thus

$$\mathbb{P}(L_a^{\diamond} > t) = 2\Phi(-a\sqrt{t}), \tag{4.9}$$

which is equivalent to (4.3). Finally,

$$\mathbb{P}(G^2/a^2 \leq t) = \mathbb{P}(|G| \leq a\sqrt{t}) = 2\Phi(a\sqrt{t}) - 1,$$

proving (4.4). □

Remark 4.2 (Time inversion). Lemma 4.1 also follows immediately from Brownian time inversion and the reflection principle. If $\widetilde{B}(u) = uB(1/u)$, then $\widetilde{B}$ is Brownian motion, and

$$\{L_a^> \le t\} = \left\{ \sup_{0 \le u \le 1/t} \widetilde{B}(u) \le a \right\}.$$

The reflection principle then gives $2\Phi(a\sqrt{t}) - 1$. See [RY99, BS02] for classical formulations of these identities.

5 Exact stabilization and independent marking

Theorem 5.1 (Exact product marking)**.** *Let I be finite or countable, let X have an arbitrary probability law on $\{0,1\}^I$, and let $(B_i)_{i\in I}$ be independent standard Brownian motions independent of X. Define Y, $\widehat{X}$, and T_i by (2.3), (2.4), and (2.5). Then every T_i is finite almost surely and*

$$\mathcal{L}(X, (T_i)_{i\in I}) = \mathcal{L}(X) \otimes \bigotimes_{i\in I} \mathrm{Gamma}\left(\frac{1}{2}, 8\right). \tag{5.1}$$

Equivalently, conditionally on the entire configuration X, the variables T_i are independent and

$$T_i \overset{\mathrm{d}}{=} 4G_i^2, \qquad G_i \sim N(0,1). \tag{5.2}$$

In particular, for every finite $A \subseteq I$ and every family $(t_i)_{i\in A}$ of nonnegative numbers,

$$\mathbb{P}(T_i \le t_i \text{ for all } i \in A \mid X) = \prod_{i\in A} \left[1 - 2\Phi\left(-\frac{\sqrt{t_i}}{2}\right)\right] \quad \text{almost surely.} \tag{5.3}$$

Proof. Condition on X and define

$$\widetilde{B}_i(t) = (1 - 2X_i)B_i(t). \tag{5.4}$$

For each fixed value of X, this is only a deterministic coordinatewise sign change. Product Wiener measure is invariant under these sign changes. Hence, conditionally on X, the paths $\widetilde{B}_i$ are independent standard Brownian motions, with a conditional law not depending on X. It follows in particular that the entire path field $(\widetilde{B}_i)_{i\in I}$ is independent of X.

The signed margin is

$$R_i(t) = (1 - 2X_i)(Y_i(t) - t/2) = \widetilde{B}_i(t) - t/2. \tag{5.5}$$

If $X_i = 0$, then coordinate i is decoded incorrectly exactly when $R_i(t) > 0$. If $X_i = 1$, it is decoded incorrectly exactly when $R_i(t) \ge 0$. Thus, conditionally on X, T_i is either $L_{1/2}^>$ or $L_{1/2}^\ge$ applied to the independent path $\widetilde{B}_i$. Lemma 4.1 shows that both choices have the same distribution,

$$T_i \overset{\mathrm{d}}{=} \frac{G_i^2}{(1/2)^2} = 4G_i^2.$$

The conditional joint law is therefore the same product law for every value of X. This proves independence from X, coordinate independence, and (5.3). Since $4G_i^2$ is gamma with shape $1/2$ and scale 8, (5.1) follows. □

Corollary 5.2 (OpenAI sampler)**.** *Under the hypotheses and notation of Proposition 2.1, set* $X = \Psi(W)$ *and define the continuous-time readout*

$$\widehat{X}_i(t) = \mathbf{1}\{Z_i(W)(t) > t/2\}. \tag{5.6}$$

The last-error field of (5.6) *satisfies* (5.1)*. In particular, this holds for the free uniform spanning forest sampler of [Ope26].*

Proof. This is the transfer of Theorem 5.1 through the joint path-law identity (2.2). □

Remark 5.3 (Why the midpoint is exact). For a threshold αt with $0 < \alpha < 1$, the two target bits yield different drift magnitudes:

$$T_i \mid \{X_i = 0\} \overset{\mathrm{d}}{=} \frac{G^2}{\alpha^2}, \qquad T_i \mid \{X_i = 1\} \overset{\mathrm{d}}{=} \frac{G^2}{(1-\alpha)^2}. \tag{5.7}$$

Only at $\alpha = 1/2$ do these conditional laws coincide. The independence in (5.1) is therefore a genuine midpoint symmetry.

6 Finite-set laws and terminal defects

Define

$$F(t) = 2\Phi(\sqrt{t}/2) - 1, \qquad q(t) = 1 - F(t) = 2\Phi(-\sqrt{t}/2). \tag{6.1}$$

Corollary 6.1 (Exact finite-set stabilization)**.** *For a finite* $A \subseteq I$ *with* $|A| = M$*,*

$$\mathbb{P}(T_A \le t \mid X) = \mathbb{P}(T_A \le t) = F(t)^M, \qquad t \ge 0. \tag{6.2}$$

The one-coordinate density and Laplace transform are

$$f_T(t) = \frac{e^{-t/8}}{\sqrt{8\pi t}}, \quad t > 0, \qquad \mathbb{E}e^{-sT} = (1+8s)^{-1/2}, \quad s \ge 0. \tag{6.3}$$

Consequently,

$$\mathbb{E}T = 4, \qquad \mathrm{Var}(T) = 32. \tag{6.4}$$

The maximum T_A *has density*

$$f_{T_A}(t) = MF(t)^{M-1}\frac{e^{-t/8}}{\sqrt{8\pi t}}, \qquad t > 0. \tag{6.5}$$

Proof. Equation (6.2) follows from Theorem 5.1. The remaining formulas follow by differentiating F and using the standard gamma transform. □

At time t, call coordinate i a *remaining terminal defect* if it will be decoded incorrectly at some time strictly greater than t. Equivalently, it is a defect when $T_i > t$.

Corollary 6.2 (Exact defect count)**.** *For finite* $A \subseteq I$*, define*

$$N_A(t) = \sum_{i \in A} \mathbf{1}\{T_i > t\}. \tag{6.6}$$

Conditionally on X*, and hence also unconditionally,*

$$N_A(t) \sim \mathrm{Bin}(|A|, q(t)). \tag{6.7}$$

In contrast, the number of coordinates decoded incorrectly at the single snapshot time $t > 0$ *is binomial with parameter* $q(t)/2$*.*

Proof. The first assertion follows from the independent marking theorem. For a snapshot error, either target bit requires a centered normal fluctuation of magnitude at least $t/2$, so the conditional error probability is $\Phi(-\sqrt{t}/2) = q(t)/2$. Conditional independence follows from the independence of the Brownian coordinates. □

Thus the factor of two in (1.2) is exact: among the paths that will err after time t, half are already in error at time t, while half are currently correct and later return across the threshold.

Corollary 6.3 (Configuration-selected coordinates)**.** *Let $A = A(X)$ be an almost surely finite subset of I selected measurably from the sampled configuration. Then*

$$\mathbb{P}(T_{A(X)} \le t \mid X) = F(t)^{|A(X)|} \quad \textit{almost surely.} \tag{6.8}$$

More generally, conditionally on X, the marks attached to the selected coordinates remain independent $\mathrm{Gamma}(1/2, 8)$ *variables.*

Proof. Condition on X and apply the product law in Theorem 5.1 to the now deterministic finite set $A(X)$. □

6.1 Exact confidence horizons

Corollary 6.4 (Exact and elementary horizons)**.** *Fix $M \ge 1$ and $0 < \delta < 1$. The unique time $t_{M,\delta}$ satisfying $\mathbb{P}(T_A \le t_{M,\delta}) = 1 - \delta$ for $|A| = M$ is*

$$t_{M,\delta} = 4\left[\Phi^{-1}\left(\frac{1 + (1-\delta)^{1/M}}{2}\right)\right]^2. \tag{6.9}$$

The simpler sufficient condition

$$t \ge 8\log(M/\delta) \tag{6.10}$$

implies $\mathbb{P}(T_A \le t) \ge 1 - \delta$.

Proof. Equation (6.9) is obtained by solving $F(t)^M = 1 - \delta$. The Gaussian Chernoff bound $2\Phi(-x) \le e^{-x^2/2}$ for $x \ge 0$ gives $q(t) \le e^{-t/8}$. Therefore

$$\mathbb{P}(T_A > t) \le Mq(t) \le Me^{-t/8},$$

which proves (6.10). □

7 Extreme stabilization times

For $M \ge 2$, let

$$b_M = 8\log M - 4\log\log M - 4\log\pi. \tag{7.1}$$

All logarithms are natural.

Theorem 7.1 (Gumbel law and Poisson terminal defects)**.** *Let $T_1, \dots, T_M$ be the stabilization marks on any M prescribed coordinates and let $T_M^{\max} = \max_{1 \le i \le M} T_i$. Then, for every $x \in \mathbb{R}$,*

$$\mathbb{P}\left(\frac{T_M^{\max} - b_M}{8} \le x\right) \longrightarrow \exp(-e^{-x}). \tag{7.2}$$

At the same scale, the remaining terminal-defect count satisfies

$$N_M(b_M + 8x) \xrightarrow{\mathrm{d}} \mathrm{Pois}(e^{-x}), \tag{7.3}$$

where $N_M(t) = \sum_{i=1}^{M} \mathbf{1}\{T_i > t\}$.

The convergence in (7.2) *holds in every fixed moment. In particular,*

$$\mathbb{E}T_M^{\max} = b_M + 8\gamma + o(1), \tag{7.4}$$

$$\mathrm{Var}(T_M^{\max}) = \frac{32\pi^2}{3} + o(1), \tag{7.5}$$

where γ *is Euler's constant.*

Proof. The normal-tail expansion gives

$$q(t) = \sqrt{\frac{8}{\pi t}} e^{-t/8} \left(1 - \frac{4}{t} + O(t^{-2})\right) \qquad (t \to \infty). \tag{7.6}$$

For fixed x, put $t_M(x) = b_M + 8x$. From (7.1),

$$Me^{-b_M/8} = \sqrt{\pi \log M}. \tag{7.7}$$

Combining (7.6) and (7.7) yields

$$Mq(t_M(x)) \longrightarrow e^{-x}. \tag{7.8}$$

Since $q(t_M(x)) \to 0$, the exact formula

$$\mathbb{P}(T_M^{\max} \le t_M(x)) = (1 - q(t_M(x)))^M$$

gives (7.2). The exact binomial law

$$N_M(t_M(x)) \sim \mathrm{Bin}(M, q(t_M(x)))$$

and (7.8) give (7.3).

For completeness, we verify the uniform integrability needed for moment convergence. Mills-ratio bounds imply constants $0 < c < C < \infty$ such that

$$c\frac{e^{-t/8}}{\sqrt{1+t}} \le q(t) \le C\frac{e^{-t/8}}{\sqrt{1+t}}, \qquad t \ge 0. \tag{7.9}$$

Write $V_M = (T_M^{\max} - b_M)/8$. For $x \ge 0$ and all sufficiently large M,

$$\mathbb{P}(V_M > x) \le Mq(b_M + 8x) \le C_1 e^{-x}. \tag{7.10}$$

For $0 \le x \le b_M/16$, the lower bound in (7.9) gives

$$\mathbb{P}(V_M < -x) \le \exp\{-Mq(b_M - 8x)\} \le \exp\{-c_1 e^x\}. \tag{7.11}$$

For $b_M/16 \le x \le b_M/8$, monotonicity gives

$$\mathbb{P}(V_M < -x) \le \exp\{-Mq(b_M/2)\} \le \exp\{-c_2 M^{1/2}/\sqrt{\log M}\}. \tag{7.12}$$

For $x > b_M/8$, the probability is zero because stabilization times are nonnegative. The bounds (7.10)–(7.12) imply uniform integrability of every fixed power of $|V_M|$. Hence the moments converge to those of a standard Gumbel random variable, whose mean and variance are γ and $\pi^2/6$, respectively. This proves (7.4)–(7.5). □

Remark 7.2 (Interpretation of the centering). The term $8\log M$ comes from the exponential factor $e^{-t/8}$ in the gamma tail. The correction $-4\log\log M$ comes from its $t^{-1/2}$ prefactor, and $-4\log\pi$ fixes the limiting Poisson intensity to be e^{-x}.

8 Unequal rates and deterministic clocks

The product marking persists when different coordinates are observed at different deterministic speeds.

Proposition 8.1 (Unequal constant rates)**.** *Let $r_i > 0$ and consider*

$$Y_i(t) = r_i t X_i + \sqrt{r_i}\, B_i(t), \qquad \widehat{X}_i(t) = \mathbf{1}\{Y_i(t) > r_i t/2\}. \tag{8.1}$$

Let $T_i^{(r)}$ be the corresponding last-error time. Conditionally on X, the variables $T_i^{(r)}$ are independent and

$$T_i^{(r)} \overset{\mathrm{d}}{=} \frac{4}{r_i} G_i^2. \tag{8.2}$$

Therefore, for finite $A \subseteq I$,

$$\mathbb{P}\Big(T_i^{(r)} \le t_i \text{ for all } i \in A \mid X\Big) = \prod_{i \in A} \left[1 - 2\Phi\left(-\frac{\sqrt{r_i t_i}}{2}\right)\right]. \tag{8.3}$$

The conditional law again does not depend on X.

Proof. After the sign change from Theorem 5.1, divide the signed margin by $\sqrt{r_i}$. The result is standard Brownian motion with drift $-\sqrt{r_i}/2$. Lemma 4.1 gives (8.2) and hence (8.3). □

Proposition 8.2 (Deterministic observation clocks)**.** *For each i, let $\kappa_i : [0,\infty) \to [0,\infty)$ be continuous, strictly increasing, unbounded, and satisfy $\kappa_i(0) = 0$. Consider*

$$Y_i(t) = \kappa_i(t) X_i + B_i(\kappa_i(t)), \qquad \widehat{X}_i(t) = \mathbf{1}\{Y_i(t) > \kappa_i(t)/2\}. \tag{8.4}$$

Then, conditionally on X, the physical last-error times are independent and satisfy

$$T_i^{(\kappa)} \overset{\mathrm{d}}{=} \kappa_i^{-1}(4G_i^2). \tag{8.5}$$

Consequently,

$$\mathbb{P}\Big(T_i^{(\kappa)} \le t_i \text{ for all } i \in A \mid X\Big) = \prod_{i \in A} \left[1 - 2\Phi\left(-\frac{\sqrt{\kappa_i(t_i)}}{2}\right)\right]. \tag{8.6}$$

Proof. In intrinsic time $u = \kappa_i(t)$, the model is exactly the unit-rate model of Theorem 5.1. Applying the inverse deterministic clock gives (8.5) and (8.6). □

8.1 A quantitative terminal-profile transfer

For a finite heterogeneous array, set

$$q_{n,i}(t) = 2\Phi\left(-\frac{\sqrt{r_{n,i} t}}{2}\right), \qquad \Lambda_n(t) = \sum_i q_{n,i}(t), \qquad q_n^*(t) = \max_i q_{n,i}(t). \tag{8.7}$$

Proposition 8.3 (Quantitative Poisson transfer)**.** *Let $T_n^{\max}$ be the maximum stabilization time in a finite array with rates $r_{n,i}$. For every t with $q_n^*(t) < 1$,*

$$0 \le -\log \mathbb{P}(T_n^{\max} \le t) - \Lambda_n(t) \le \frac{\sum_i q_{n,i}(t)^2}{2(1 - q_n^*(t))}. \tag{8.8}$$

In particular, if for every fixed x,

$$q_n^*(t_n(x)) \to 0, \qquad \Lambda_n(t_n(x)) \to \Lambda(x) < \infty, \tag{8.9}$$

then

$$\mathbb{P}(T_n^{\max} \le t_n(x)) \longrightarrow e^{-\Lambda(x)}. \tag{8.10}$$

At the same time, the number of coordinates with a future error after $t_n(x)$ converges to $\mathrm{Pois}(\Lambda(x))$.

Proof. The exact product formula gives

$$\mathbb{P}(T_n^{\max} \le t) = \prod_i (1 - q_{n,i}(t)).$$

For $0 \le z < 1$,

$$0 \le -\log(1-z) - z = \sum_{k \ge 2} \frac{z^k}{k} \le \frac{z^2}{2(1-z)}. \tag{8.11}$$

Summing (8.11) proves (8.8). Under (8.9),

$$\sum_i q_{n,i}(t_n(x))^2 \le q_n^*(t_n(x)) \Lambda_n(t_n(x)) \to 0,$$

which proves (8.10). The Poisson convergence is the standard law-of-small-numbers consequence for independent Bernoulli defects; Le Cam's inequality gives total-variation error at most $\sum_i q_{n,i}(t_n(x))^2$ [LC60]. □

9 Boundary, filtration, and infinite-coordinate issues

9.1 The final threshold contact

Almost surely, the signed margin in (5.5) equals zero at its last boundary contact. With the strict decoder convention, a target-0 coordinate is correct at that contact, whereas a target-1 coordinate is incorrect. Definition 2.2 avoids any ambiguity: every time strictly greater than T_i is correct, and the strict and non-strict last-passage functionals have the same law by Lemma 4.1.

9.2 The last-error time is not a stopping time

Let $R(t) = B(t) - t/2$ and let T be its last visit to the nonnegative half-line (or its last positive time; the conditional formula is the same). With $\mathcal{F}_t = \sigma(R(s) : 0 \le s \le t)$, the hitting formula (4.5) gives

$$\mathbb{P}(T \le t \mid \mathcal{F}_t) = \mathbf{1}\{R(t) < 0\}(1 - e^{R(t)}). \tag{9.1}$$

This conditional probability takes values strictly between 0 and 1 with positive probability. Hence $\{T \le t\}$ is not $\mathcal{F}_t$-measurable, and T is not a stopping time. The exact horizons in Corollary 6.4 are therefore probabilistic certification times, not pathwise online certificates.

9.3 No common finite time on infinitely many equal-rate coordinates

If I is countably infinite and all coordinates have unit rate, then every T_i is finite almost surely, but

$$\sup_{i \in I} T_i = \infty \quad \text{almost surely.} \tag{9.2}$$

Indeed, for every finite t, $F(t) < 1$, so the probability that all infinitely many independent stabilization marks are at most t is zero. Taking a countable union over integer t proves (9.2).

9.4 Continuous and discrete inspection are different

The law obtained here concerns the continuous-time readout $t \mapsto \mathbf{1}\{Z_i(t) > t/2\}$. If one inspects the sampler only at integer times, the last observed error is bounded above by T_i but has a different lattice-valued distribution. The asymptotic slope readout in [Ope26] needs only integer times; the exact stabilization law uses the full reconstructed continuous path.

10 Concluding remarks

The OpenAI construction supplies an exact observation representation, while the midpoint rule turns the two target states into the same negative-drift Brownian last-passage problem. This yields more than a marginal tail estimate: it produces an independent gamma marking of the entire sampled configuration. For a finite family, permanent stabilization is the zero-defect event in an exact binomial process. The Gumbel law is consequently the zero-count limit of an explicit Poisson terminal-defect field.

The result is insensitive to the dependence structure of the sampled binary process. All model-specific work is confined to establishing the Brownian sampler representation. Once that representation holds, the stabilization law, its extreme-value limit, and its deterministic-clock extensions are universal.

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